

Hasan Khaddour

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ABOUT ME

A 22-year-old **Software and Artificial Intelligence** Engineer from the Higher Institute for Applied Sciences and Technology (HIAST). With two years of experience in AI, web development, and software architecture, skilled in solving complex problems and adaptable to various work environments. Passionate about designing and developing scalable software solutions while leveraging the latest advancements in AI.

EDUCATION

BS in Software and AI Engineering, Higher Institute for Applied Sciences and Technology (HIAST) 10/2020 - 08/2025
Damascus, Syria

- Website: ([Link](#))

PROJECTS

Event Detection from Arabic Social Media Using Deep Learning Techniques 04/2025 – 08/2025

- Architected a **document-pivot event detection pipeline** tailored for **Arabic social media streams** (Twitter, Telegram, news), addressing the scarcity of domain-specific resources. ([Link](#))
- Curated and annotated a **20K-tweet Arabic dataset** (SAS Golden, Labeled, Silver), providing a benchmark resource for event detection in low-resource languages.
- Built an **incremental clustering framework** leveraging **mBERT embeddings**, **graph neural networks**, and **contrastive learning**, with periodic retraining to adapt to evolving discourse.
- Enhanced system reliability with **binary filtering classifiers** (SVM, KNN) and **LLM-based abstractive summarization**, producing concise, ranked event summaries.
- Implemented a **location inference module** using CaMel NER, and heuristics to enrich events with geospatial context.
- Achieved robust performance on real-world May 2025 events dataset, evaluated with **NMI**, **AMI**, **ARI** clustering metrics.
- **Tech stack**: Python, PyTorch, PyTorch Geometric, HuggingFace Transformers, Scikit-learn, NLTK/Stanza, Flask.

Situation Awareness System (SAS) 04/2025 – 08/2025

- Designed and delivered a **scalable microservices-based platform** for monitoring and visualization of events on an interactive map. ([Link](#))
- Engineered modular services including **scrapping**, **event detection**, **summarization**, **location inference**, and **identity management**, orchestrated via Kafka and exposed through a secure gateway.
- Implemented a **resilient backend** with ASP.NET Core and Flask services, supported by **SQL Server**, **MongoDB**, and event-driven processing pipelines.
- Built an **Angular-based frontend** with modular vertical slices and interactive geospatial visualization (Leaflet.js).
- Ensured scalability and reliability through **containerization (Docker)**, CI/CD pipelines (Jenkins), and load testing (k6).
- **Tech stack**: ASP.NET Core, Angular, Flask (Python), Kafka, SQL Server, MongoDB, IdentityServer, PrimeNG, Leaflet.js, Docker, Jenkins, YARP, JWT, Grafana k6.

Sentiment Classification of Mental Health Discourse 10/2024 – 12/2024

- Developed a machine learning pipeline on a curated 50K mental health dataset, performing preprocessing, feature engineering, and sentiment classification. ([Link](#))
- Designed a **two-stage framework** — first detecting Normal vs. Non-Normal discourse, then applying multi-class classification for finer-grained sentiment categories.
- Benchmarked traditional ML models BERT-based embeddings, deriving **domain-specific insights** relevant for early detection and monitoring of mental health discourse online.
- **Tech stack**: Python (Scikit-learn, Pandas, SciPy), NLP (NLTK, spaCy, Wod2Vec, BERT), Deep Learning (TensorFlow/Keras).

Projects Management (Planning & Tracking) Web Application – HIAST 07/2024 – 10/2024

- Designed and implemented a **full-stack web application** to digitize project planning, monitoring, and reporting at HIAST, replacing manual paper-based processes and improving decision-making efficiency. ([Link](#))
- Enabled **project managers and team members** to track progress and monitor individual contributions.
- Applied **Clean Architecture** with **Domain-Driven Design (DDD)** and design patterns in ASP.NET Core, ensuring scalability, flexibility, and long-term maintainability.
- Developed a modular, responsive UI with **Angular**, seamlessly integrated with RESTful backend APIs.
- **Tech Stack**: ASP.NET Core, Entity Framework Core, Angular, SQL Server, REST APIs.

eMedic – Prescriptions and Medicines Management System05/2024 – 07/2024

- Designed and implemented a web-based system to digitize prescriptions and manage pharmacy inventory, reducing manual errors.
- Applied Clean Architecture principles with ASP.NET Core, enabling maintainability and scalability. (Link)
- Tech Stack: ASP.NET Core, Entity Framework Core, SQL Server, Bootstrap.

EXPERIENCES

Freelance Web Developer2023 – Present

- Delivered scalable, production-grade web applications, applying Clean Architecture and DDD in ASP.NET Core.
- Built and deployed modular frontends in React.js and Angular, enhancing user experience and maintainability.

International Collegiate Programming Contest (ICPC) Contest – HIAST2021, 2022, 2024

- Participated in the HIAST-CPC, showcasing algorithmic problem-solving and teamwork. (Link)

Syrian Informatics Olympiad – Syria2018, 2019

- Competed in the Syrian Scientific Olympiad in Informatics at the national level, demonstrating advanced problem-solving and analytical skills.

CERTIFICATIONS

RISE MICCAI Online Summer School 2025 – Medical AI07/2025

- Participated via the local hub at HIAST, Damascus, Syria (July 14–18, 2025).
- Gained hands-on experience in MRI classification, hippocampus segmentation, and ensemble methods.

Software Design and ArchitectureFrom University of Alberta through Coursera01/2025

Agile Project ManagementFrom Google through Coursera02/2025

Machine LearnningFrom DeepLearning.AI through Coursera07/2024

Meta Frontend DeveloperFrom Meta through Coursera08/2023

Soft Skills

- Problem Solving & Critical Thinking
- Communication & Collaboration
- Adaptability & Continuous Learning
- Leadership & Initiative
- Time Management & Organization
- Attention to Detail & Code Quality

Technical SKILLS

Programming Languages: C + +, C, C#, Python, Java, JavaScript.

Software Architecture: Clean Architecture, Domain-Driven Design (DDD), MVC, Vertical Slice Architecture.

Software Engineering Principles: Design Patterns, SOLID Principles, Scalability & Maintainability.

Backend Development: ASP.NET Core, Flask, SQL Server, Oracle Database, Entity Framework Core, RESTful APIs, CQRS.

Frontend Development: React.js, Angular, Three.js, D3.js, Modular UI Design.

Data Science / AI / ML: Python ML libraries (Scikit-learn, PyTorch, TensorFlow), HuggingFace Transformers, PyTorch Geometric (GNNs), NLP (Stanza, NLTK, CaMel NER), Contrastive Learning.

Tools & DevOps: Git, GitLab, Jenkins, Docker, Kafka, YARP, JWT, Grafana k6.

LANGUAGE SKILLS

Mother tongue: Arabic

Other languages:

	Understanding		Speaking		Writing
	Listening	Reading	Spoken production	Spoken Interaction	
English	C1	B2	B1	B1	B2
French	B1	B2	B1	B1	B1